

A Structural models and population states (solid = read-supported; dashed = inference)

retained state (majority of smbu08 reads, ~85%)

attL-X' (38,573 bp)—attR

excised state (~15%; the delivered smbu08 chromosome assembly)

attj (146 bp, byte-identical to attR)

excision (both states)

Plasmid2 77,437 bp: two tandem units of attL+X
boundary A (read-supported)

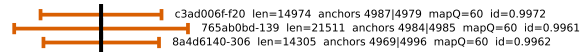


tandem-dimer representation; a monomer circle
at 2x copy number is sequence-equivalent

Read evidence: 41 reads strongly place the attj state (chr08) over the retained model;
1,883 reads span the plasmid boundary (mapQ ≥ 20); retained:excised chromosome counts ≈ 5.6:1.

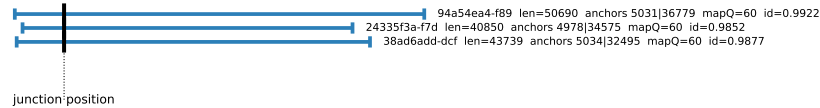
B Junction-spanning Nanopore reads (single primary alignment, mapQ ≥ 20, ≥ 1 kb anchors both sides)

attj (chr08) — loose reads: 46 strict reads: 26



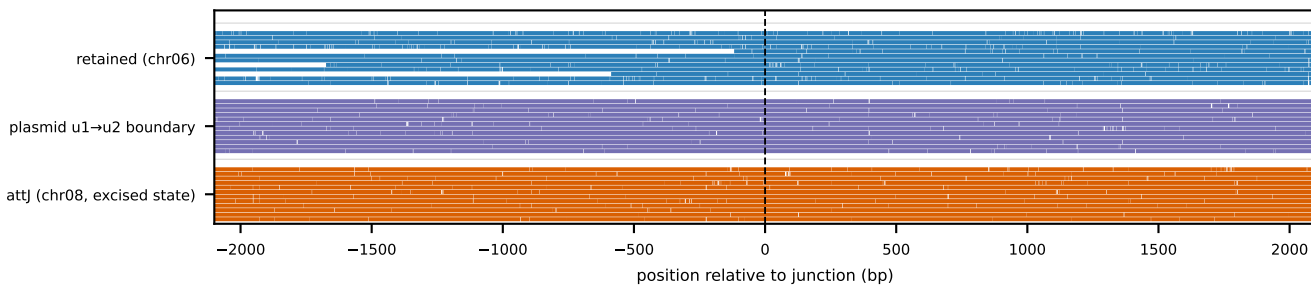
plasmid boundary (x2 on circle) — loose reads: 1,883 strict reads: 992

retained junction (chr06) — loose reads: 257 strict reads: 149

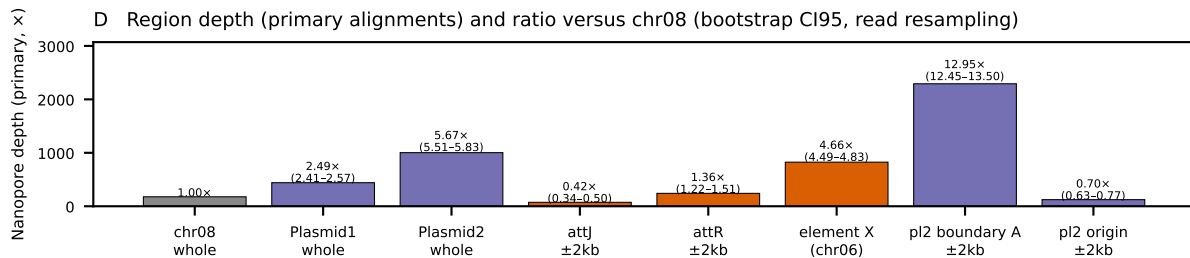


junction position

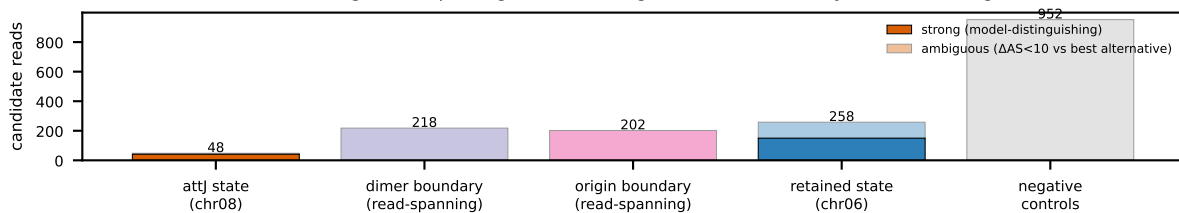
C Read alignments across key junctions (±2 kb window of the 10-kb junction references), IGV-style



D Region depth (primary alignments) and ratio versus chr08 (bootstrap CI95, read resampling)



E Per-read best-model assignment (pre-registered: strong = ΔAS ≥ 10, identity ≥ 90%, coverage ≥ 30%)



F Chromosome state counts: smbu08 contains both states (257:46 ≈ 5.6:1 retained:excised); smbu06 control is 24:1

